Understanding America's Childcare Markets

An Unsupervised Machine Learning Approach to County-Level Childcare Pricing

BA820 | Section A1 | Team 4
Shon Shaju • Akhil Nair • Hemanth Kumar Gopi • Kai Tran

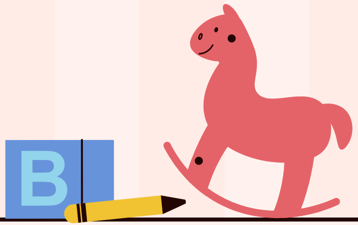
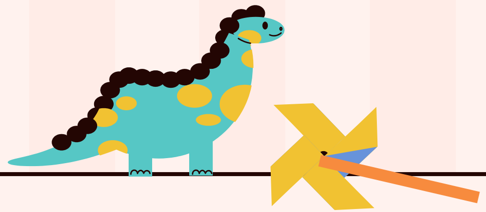


Table of contents



01

Introduction & Data Overview

Introduction

"In some counties, families spend more on childcare than on housing. Yet we treat all counties the same." — *Motivated by: U.S. Dept. of Labor Women's Bureau, 2023 Issue Brief*

Why this research matters

Cost crisis

Childcare costs range from 8% to over 20% of family income depending on county

Mothers leaving workforce

mothers in high-cost counties are quietly exiting the workforce. Not by choice, but because the math no longer works

Only 16% Get Help

Despite federal subsidies, only 16% of eligible families receive childcare assistance. States ration limited funds, and many parents don't know they qualify.

Stakeholder:

Parents

Employers

Data Overview

3,144
U.S. Counties

11
Years (2008–2018)

~34K
County-Year Rows

Features

Category	Variables
Childcare Costs (6)	Center & family-based × infant, toddler, preschool (weekly $)
Income	Median household income (2018$)
Labor Market (2)	Female labor participation, unemployment rate (ages 20–64)
Poverty	Poverty rate for individuals
Engineered (4+)	*Affordability ratio, infant–preschool gap, center–family gap, provider type ratio*

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Notes

- Missingness: ~7–8% of price columns missing, esp. rural counties.
-

02

Pipeline Overview

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If childcare markets aren't the same, what’s different?

1. Do counties form distinct childcare market types?
2. How do market types differ in affordability and maternal labor force participation?
3. Do pricing structures align more with socioeconomic similarity than geographic proximity?
4. Which counties are structurally extreme and what distinguishes them?

Turning County-Level Data into Market Segments

Raw Data

34K county-year observations

→

Preprocess

Aggregate, impute, log transform, scale

→

PCA

"One big gradient" (M2/M3)

→

UMAP

Non-linear: 6 types found (M4)

→

Cluster Profiling

Real $ values, policy narratives

Key Evolution: PCA (linear) saw a single affordability gradient; UMAP (non-linear) saw six different market types.

Preprocessing Steps:

County-level averaging (**3,144 counties**) → Median imputation (**~8% missing**) → Log$_{1+}$ transform on **10** skewed features → StandardScaler

Log transform was the key M4 fix: UMAP builds a neighborhood graph sensitive to density. Right-skewed costs distorted neighborhoods without it.

Q1: Do Counties Form Distinct Market Types?

Kai Tran

What Was Explored:

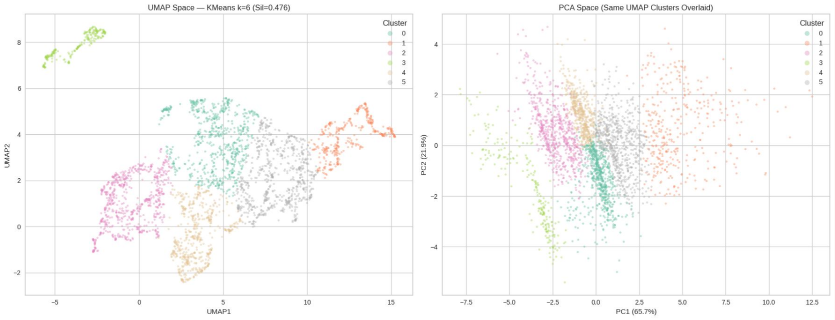
PCA + **KMeans** + Hierarchical (M2/M3): k=2 just isolated outliers. After removing top 5%, k=5 worked but **silhouette only 0.24.**

UMAP + KMeans (M4): 4×4 parameter grid tested. Best config: n_neighbors=15, min_dist=0.1. k=6 with **silhouette 0.48** — nearly double M3.

Stability: **10** random seeds → **silhouette std = 0.003**. Structure is real.

M3 vs M4

	M3 (PCA)	M4 (UMAP)
Clusters	5	6
Balance ratio	0.351	0.324
Silhouette	0.247	0.467



Takeaway: *Counties do form distinct types, but only visible through non-linear analysis. PCA saw one gradient; UMAP found 6 groups hiding inside it.*

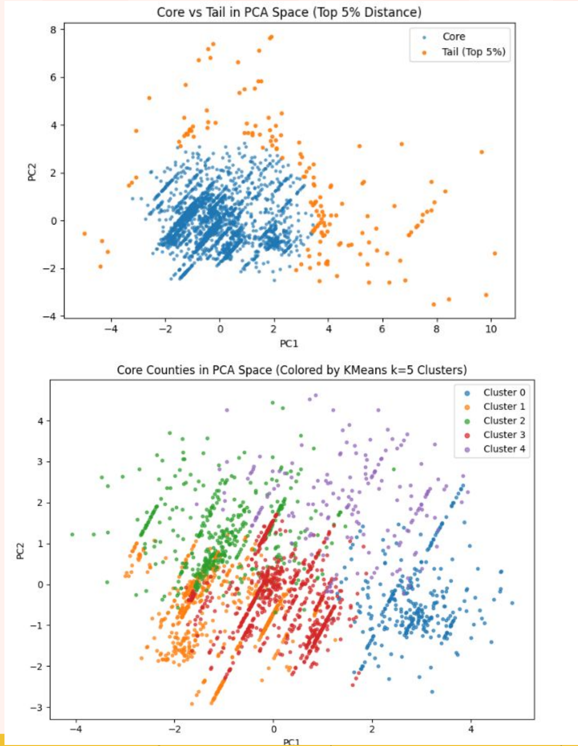
Q2: How do market types differ in affordability and maternal labor force participation?

From **M3** we learned that:

- Silhouette scores ~0.24, weak separation
- Outlier counties distorted all clusters
- No affordability context price only

For **M4:**

- Added infant/preschool ratio + center-family gap as structure features
- PCA to isolate top 5% extreme tail → re-cluster on core



Takeaway: *PCA revealed a strong overall affordability gradient, with a small group of extreme high-cost counties driving much of the apparent separation.*

Q3: Do pricing structures align more with socioeconomic similarity than geographic proximity?

What Was Explored:

From **M3** we learned that:

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- No affordability context price only

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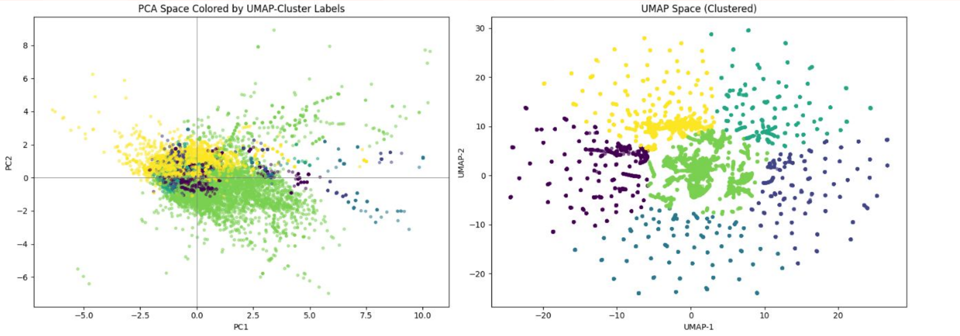

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Q4: Which counties are structurally extreme and what distinguishes them?

Hemanth

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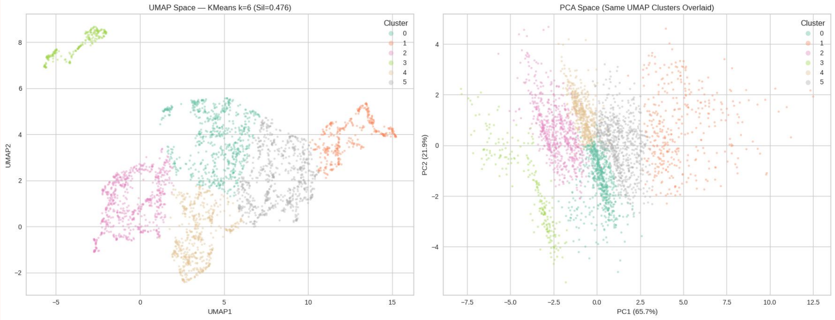

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The Integrated Picture

1

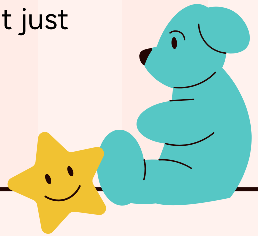
6 Market Types: Markets aren't random; they fall into 6 clear "personalities" defined by complex cost-to-income ratios

1

Economic DNA: Markets are driven by the DNA of local economics (income, poverty, unemployment), making state-line geography secondary.

Wealthy Outliers: The "Crisis" looks different in affluent cities; high wealth creates a massive supply bottleneck and price spikes, proving this isn't just a poverty issue.

Buffer Effects: Working mothers' labor choices are influenced by cultural and family support factors, not just the raw price tag of childcare.



Toys for cognitive development

Venus

Venus has a beautiful name and is the second planet from the Sun

Mercury

Mercury is the closest planet to the Sun and the smallest of them all

Mars

Despite being red, Mars is actually a cold place. It's full of iron oxide dust

Toys for group play

Mars

Mars is actually a very cold place

Venus

Venus has extremely high temperatures

Jupiter

Jupiter is the biggest planet of them all

Saturn

Saturn is a gas giant and has several rings

Awesome words

Toy quest challenge

You are in Toy Emporium, faced with the exciting task of **selecting a companion for your upcoming learning journey**. After choosing, **justify how this toy will help you:**

01 The RoboRover

An interactive robot designed for scientific adventures

02 Artistic Easel

A magical canvas that brings drawings to life

03 MathMaster Toolbox

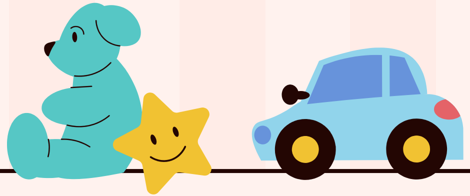
Puzzles and game to make math an enchanting experience

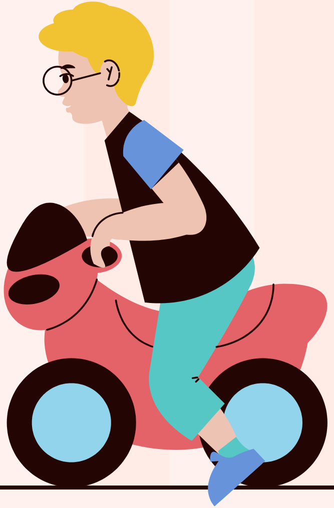
Toy chosen:

Justification:

Whoa!

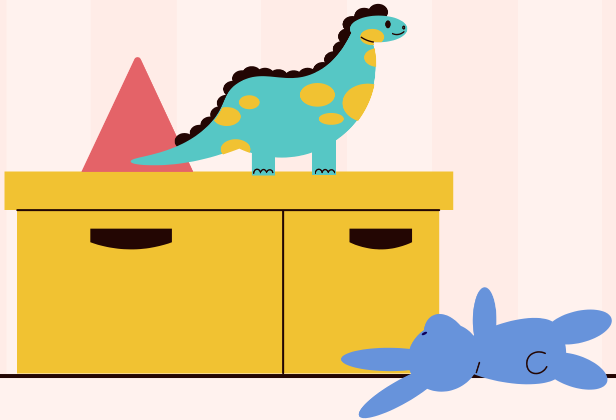
This can be the part of the presentation where you introduce yourself, write your email…





“This is a quote, words full of wisdom that someone important said and can make the reader get inspired”

—Someone Famous




A picture is worth a
thousand words

Trustworthy reviews

“Mars is actually a very cold place”

—Chris Bond

“Mercury is the closest planet to the Sun”

—Laura Lee

”Neptune is the farthest planet from the Sun”

—Rosa Sons

”Saturn is a gas giant and has several rings”

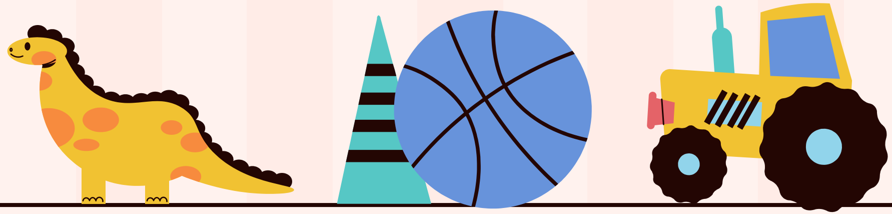
—Mary Jane

”Jupiter is the biggest planet of them all”

—John White

A picture reinforces the concept

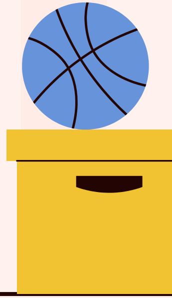
Images reveal large amounts of data, so remember: use an image instead of a long text. Your audience will appreciate it



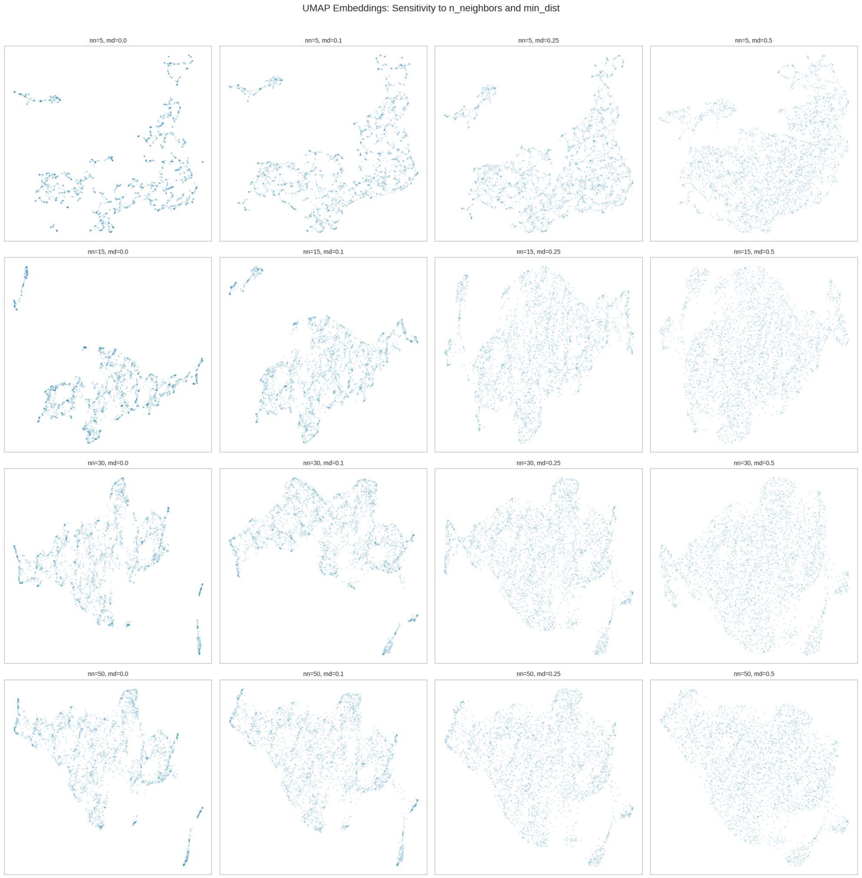
Appendix

Skewness for 10 Features

```
Skewness after log1p (county-level)
mc_infant          0.969
mc_toddler         0.899
mc_preschool       0.915
mfcc_infant        0.713
mfcc_toddler       0.572
mfcc_preschool     0.525
mhi_2018           0.358
flfpr_20to64      -0.794
unr_20to64        -0.649
pr_p              -0.202
dtype: float64

Skewness before (county-level)
mc_infant          2.148
mc_toddler         2.065
mc_preschool       1.979
mfcc_infant        1.832
mfcc_toddler       1.751
mfcc_preschool     1.814
mhi_2018           1.411
flfpr_20to64      -0.383
unr_20to64         0.974
pr_p               0.970
dtype: float64
```



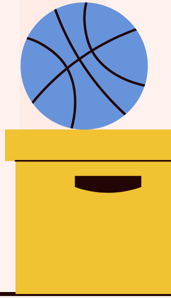
UMAP Parameter Grid



Finding best nn, min_d, k

```
best k per UMAP config (by silhouette)
 n_neighbors  min_dist  k  silhouette  balance
          15      0.10  3      0.4781    0.144
          15      0.25  2      0.4570    0.903
          30      0.10  2      0.4983    0.841
          30      0.25  2      0.4551    0.854
          50      0.10  2      0.4827    0.866

silhouette scores (config x k)
k                         2      3      4      5      6      7      8
n_neighbors min_dist
15          0.10      0.450  0.478  0.465  0.462  0.476  0.457  0.462
            0.25      0.457  0.402  0.395  0.404  0.426  0.425  0.424
30          0.10      0.498  0.430  0.407  0.442  0.478  0.464  0.455
            0.25      0.455  0.389  0.390  0.400  0.421  0.423  0.424
50          0.10      0.483  0.408  0.418  0.444  0.445  0.452  0.458
```

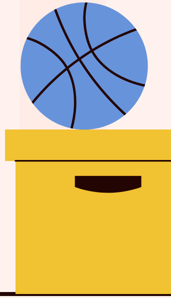
```
k=3: silhouette=0.4781, balance=0.144
0    1567
1    1352
2     225
Name: count, dtype: int64

k=5: silhouette=0.4622, balance=0.233
0    671
1    546
2    225
3    967
4    735
Name: count, dtype: int64

k=6: silhouette=0.4764, balance=0.323
0    662
1    371
2    663
3    225
4    526
5    697
Name: count, dtype: int64
```



Stability Across Seed

	seed	silhouette	balance	smallest	largest
0	0	0.4714	0.322	228	708
1	1	0.4650	0.314	228	726
2	2	0.4733	0.312	225	720
3	3	0.4729	0.332	228	686
4	4	0.4727	0.318	228	718
5	5	0.4667	0.330	228	690
6	6	0.4678	0.320	228	712
7	7	0.4700	0.332	225	677
8	8	0.4692	0.322	225	698
9	9	0.4684	0.332	228	687

```
Silhouette: mean=0.4697, std=0.0027
Range: [0.4650, 0.4733]
```



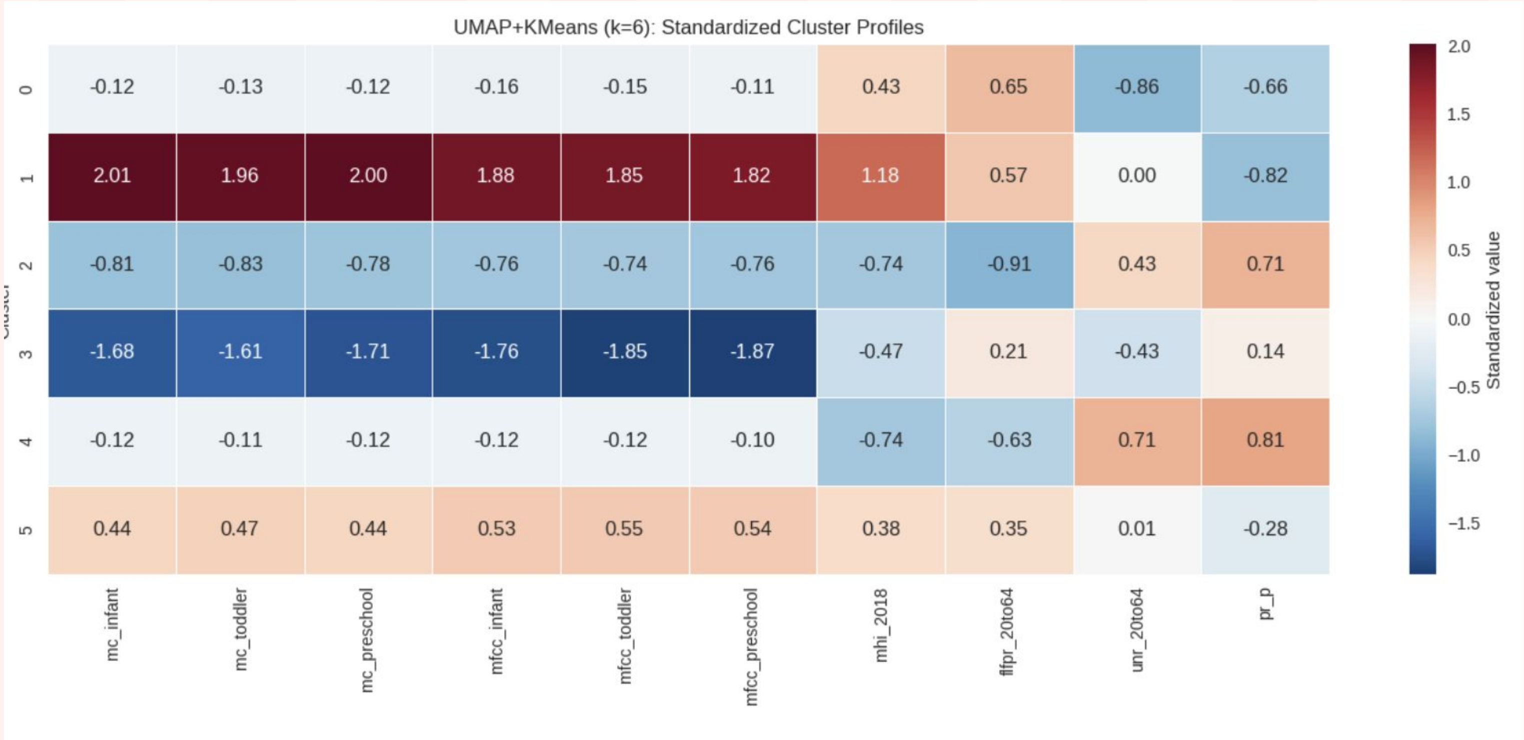
Cluster Profiles (Original Units)

cluster	mc_infant	mc_toddler	mc_preschool	mfcc_infant	mfcc_toddler	mfcc_preschool	mhi_2018	flfpr_20to64	unr_20to64	pr_p
0	-0.120	-0.133	-0.122	-0.157	-0.149	-0.108	0.434	0.647	-0.862	-0.656
1	2.010	1.961	2.001	1.877	1.846	1.824	1.182	0.566	0.002	-0.819
2	-0.805	-0.826	-0.782	-0.762	-0.744	-0.761	-0.745	-0.906	0.427	0.712
3	-1.677	-1.612	-1.707	-1.757	-1.846	-1.873	-0.472	0.209	-0.429	0.143
4	-0.122	-0.110	-0.122	-0.121	-0.115	-0.100	-0.736	-0.627	0.710	0.811
5	0.443	0.473	0.438	0.533	0.550	0.536	0.375	0.352	0.013	-0.276

cluster Means in original units

cluster	mc_infant	mc_toddler	mc_preschool	mfcc_infant	mfcc_toddler	mfcc_preschool	mhi_2018	flfpr_20to64	unr_20to64	pr_p
0	134.0	120.5	113.9	105.6	99.8	98.9	55128.6	74.9	4.6	12.0
1	230.1	195.9	180.8	159.9	148.8	144.5	66965.0	74.3	6.6	11.7
2	113.3	103.3	99.1	93.7	88.9	87.3	41554.8	63.6	8.1	20.4
3	91.4	86.7	81.4	76.9	71.8	70.8	44552.4	71.7	6.3	17.3
4	133.9	121.2	113.9	106.2	100.4	99.0	41416.9	65.3	9.0	20.9
5	154.1	138.4	128.6	121.3	114.5	111.9	54395.7	72.6	6.6	13.9

Cluster Profiles (Heatmap)



Educational toy exercises

Roll a dice to determine the toy category, then **choose a toy from the specified category.** Perform the corresponding exercise listed in the table. Discuss and learn from the educational aspects of each activity

Toy category	Dice numbers	Exercise
Building Blocks	1 and 2	Architectural Adventure: Build a city or a house, discussing the purpose of different structures.
Puzzles	3 and 4	Problem-Solving Puzzle: Work on a challenging puzzle together, emphasizing problem-solving strategies.
Science Kits	5 and 6	Mini Scientist Experiment: Choose a science kit and conduct an experiment, discussing the principles.

333,000

The Sun’s mass compared to Earth’s

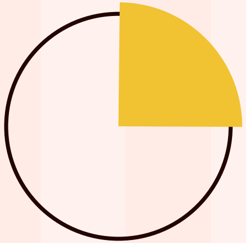
9h 55m 23s

Jupiter’s rotation period

386,000 km

Distance between Earth and the Moon

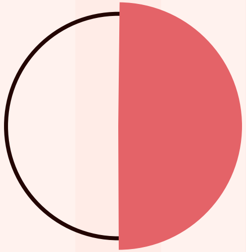
Let’s use some percentages



25%

Mercury

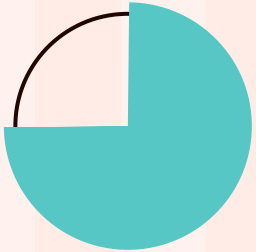
Mercury is the closest planet to the Sun and the smallest of them all



50%

Venus

Venus has a beautiful name and is the second planet from the Sun



75%

Mars

Despite being red, Mars is actually a cold place. It’s full of iron oxide dust

Computer mockup

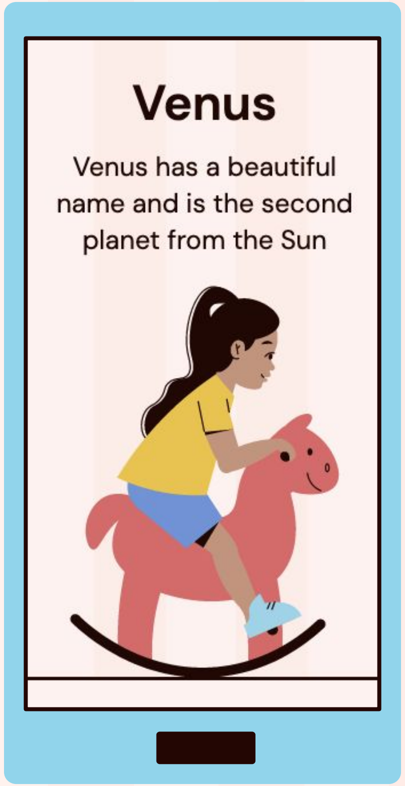
You can replace the image on the screen with your own work. Just right-click on it and select “Replace image”

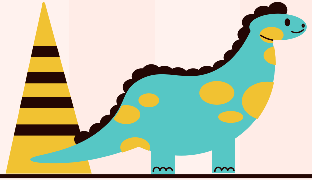
Tablet mockup

You can replace the image on the screen with your own work. Just right-click on it and select “Replace image”

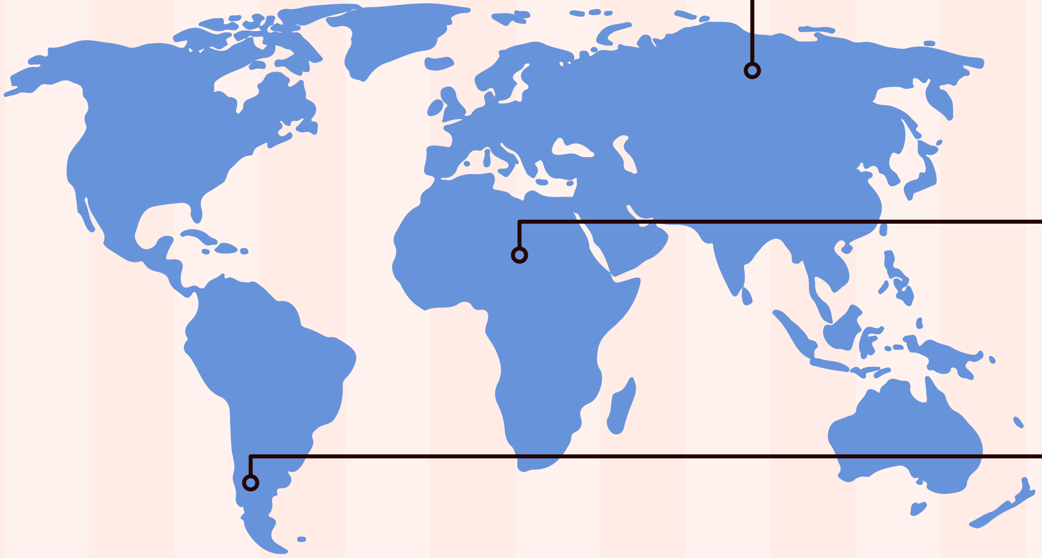
Phone mockup

You can replace the image on the screen with your own work. Just right-click on it and select “Replace image”





Educational centers



Venus

Venus is the second planet from the Sun

Mercury

Mercury is the closest planet to the Sun

Mars

Despite being red, Mars is a very cold place

03

Art and craft

You can enter a subtitle here if you need it

Important dates in toys history

Venus is the second planet from the Sun

Blocks

1890

Mercury is the closest planet

Play family

1930

Despite being red, Mars is very cold

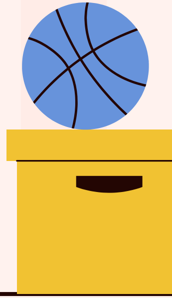
Boards

1960

Jupiter is the biggest planet

Electronic

1980



Learning through games

Mercury

Mercury is the closest planet to the Sun

Mars

Mars is a red planet

1. This is an item
2. This is an item

Jupiter

Jupiter is a gas giant

1. This is an item
2. This is an item

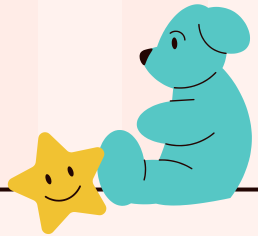
Venus

Venus is a hot planet

1. This is an item
2. This is an item

Educational toys benefits

Benefit	Examples	Age Group
Cognitive Development	Puzzles, Brain teasers, Educational games	All ages
Motor Skills	Building blocks, Art supplies, Manipulative toys	Early grades
Social Skills	Board games, Cooperative play sets, Team-building toys	Preschool
STEM Learning	Science kits, Coding toys, Engineering building sets	Elementary



Balanced use of technology

30%

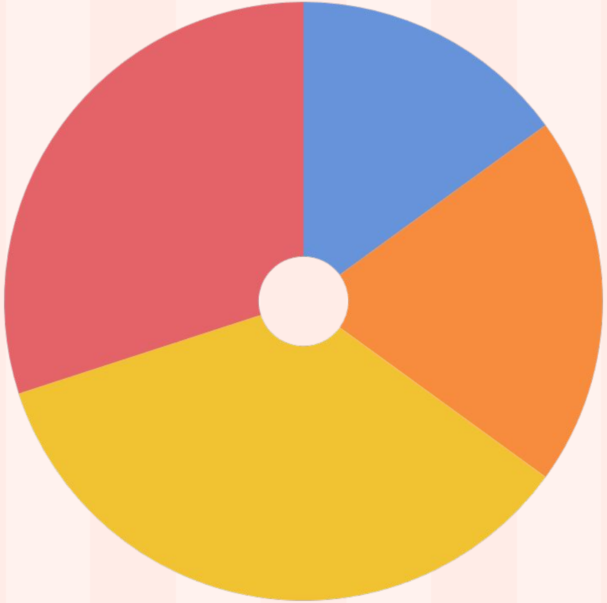
Mercury

Mercury is quite a small planet

35%

Jupiter

Jupiter is an enormous planet



15%

Venus

Venus has very high temperatures

20%

Saturn

Saturn is a gas giant with rings

Follow the link in the graph to modify its data and then paste the new one here. **For more info, click here**

Our team

Sofia Hill

You can speak a bit about this person here

Kaliyah Harris

You can speak a bit about this person here

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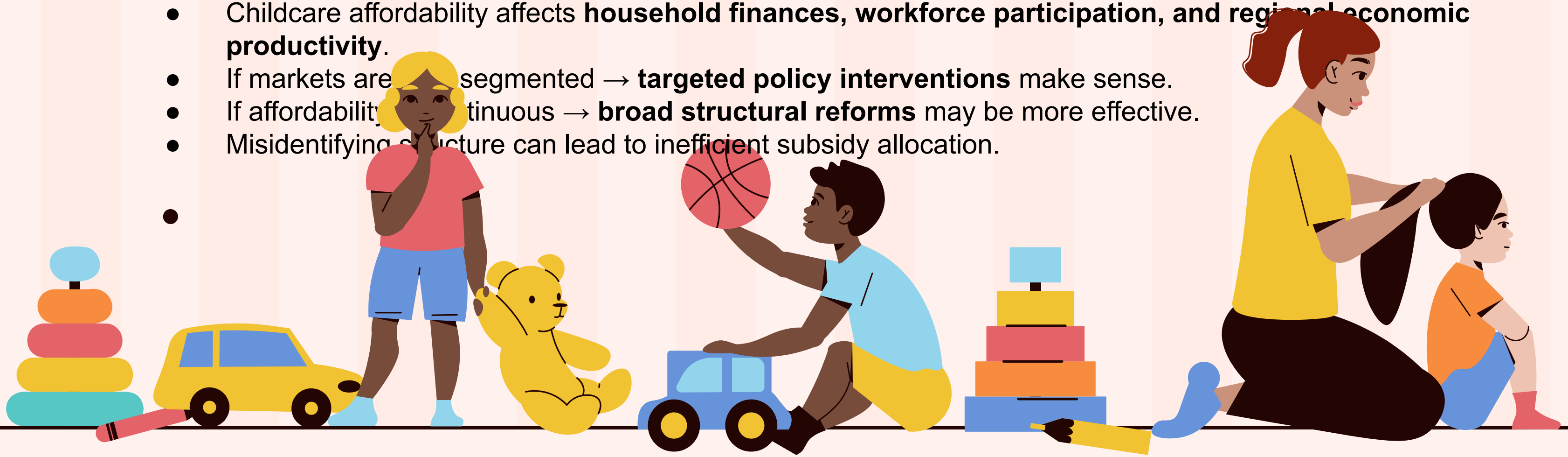
Vectors

- Kindergarten set of isolated icons with toys and characters of kids practicing with teacher playing games vector illustration
- Kindergarten infographics with characters of kids playing with toys illustration

Why cares?

- Childcare affordability affects **household finances, workforce participation, and regional economic productivity**.
- If markets are segmented → **targeted policy interventions** make sense.
- If affordability tinuous → **broad structural reforms** may be more effective.
- Misidentifying structure can lead to inefficient subsidy allocation.
-



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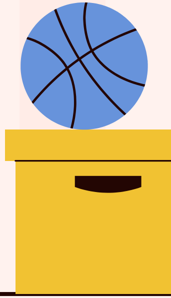
Vectors

- Kindergarten set of isolated icons with toys and characters of kids practicing with teacher playing games vector illustration
- Kindergarten infographics with characters of kids playing with toys illustration
- Flat design nursery school landing page

Photos

- Father teaching baby to play with toys at home
- Close up on child playing with didactic game
- Young woman with mobile outdoor
- Medium shot man wearing hair clip



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#e46368 #f78b3e #f1c232 #ffe8e2

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Pana

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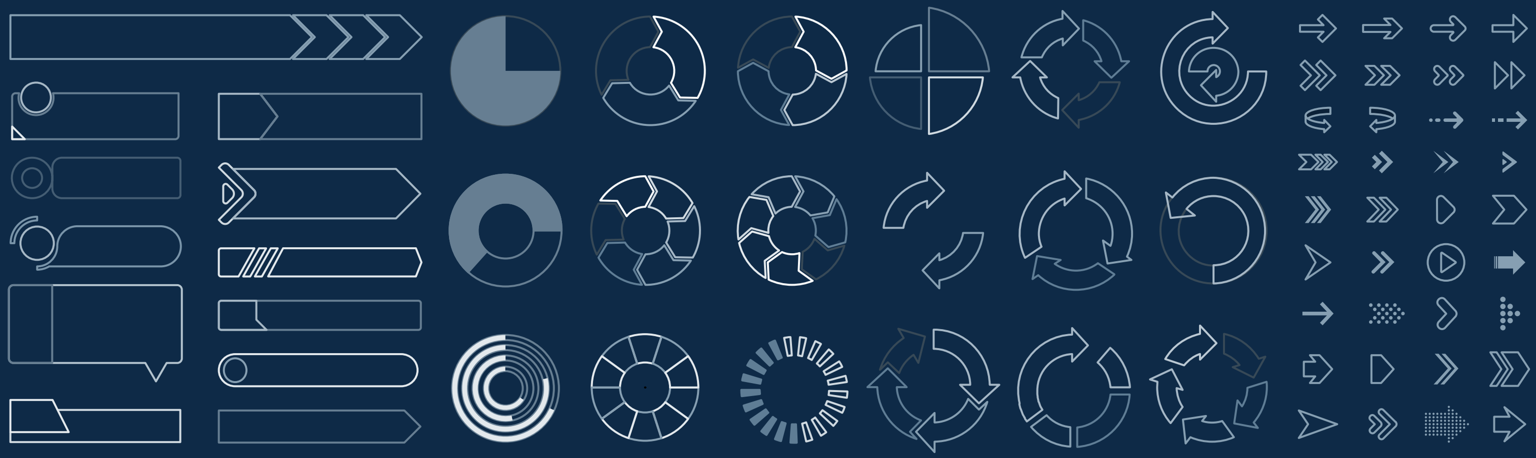







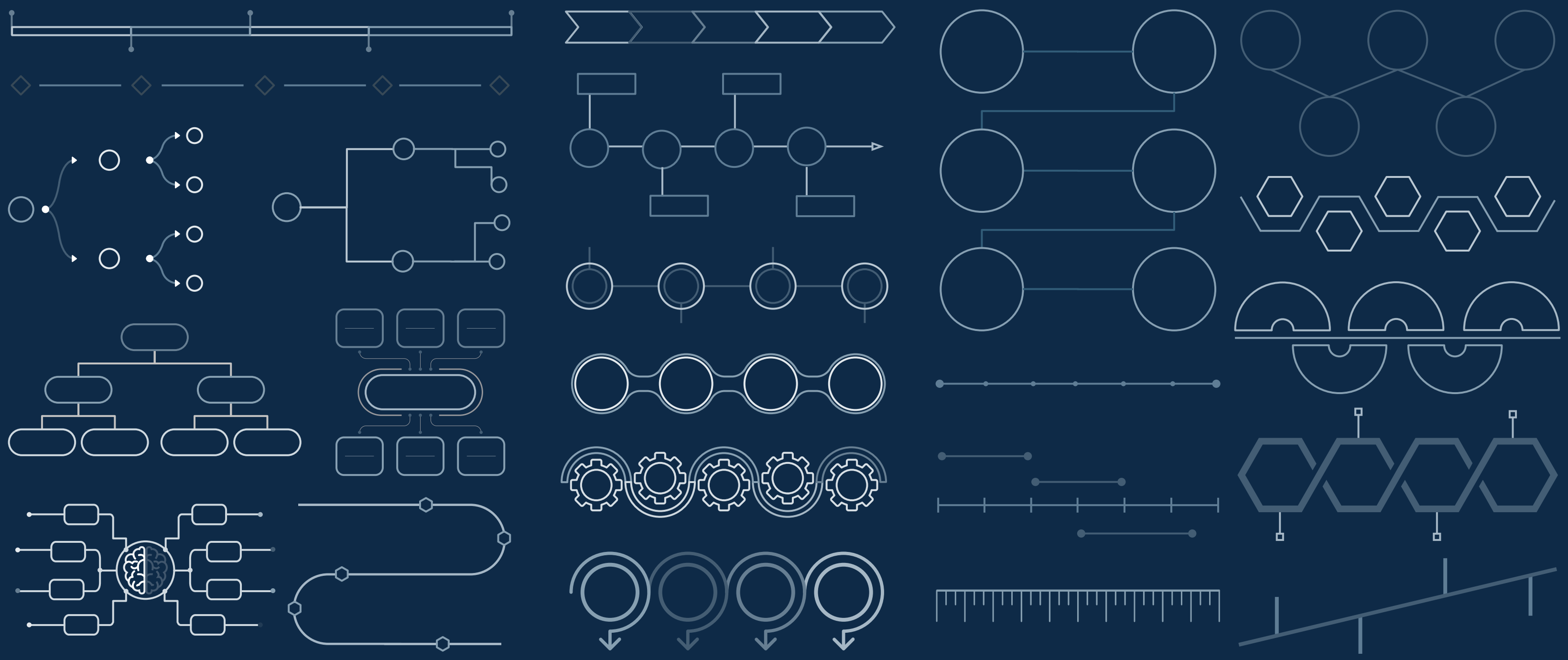

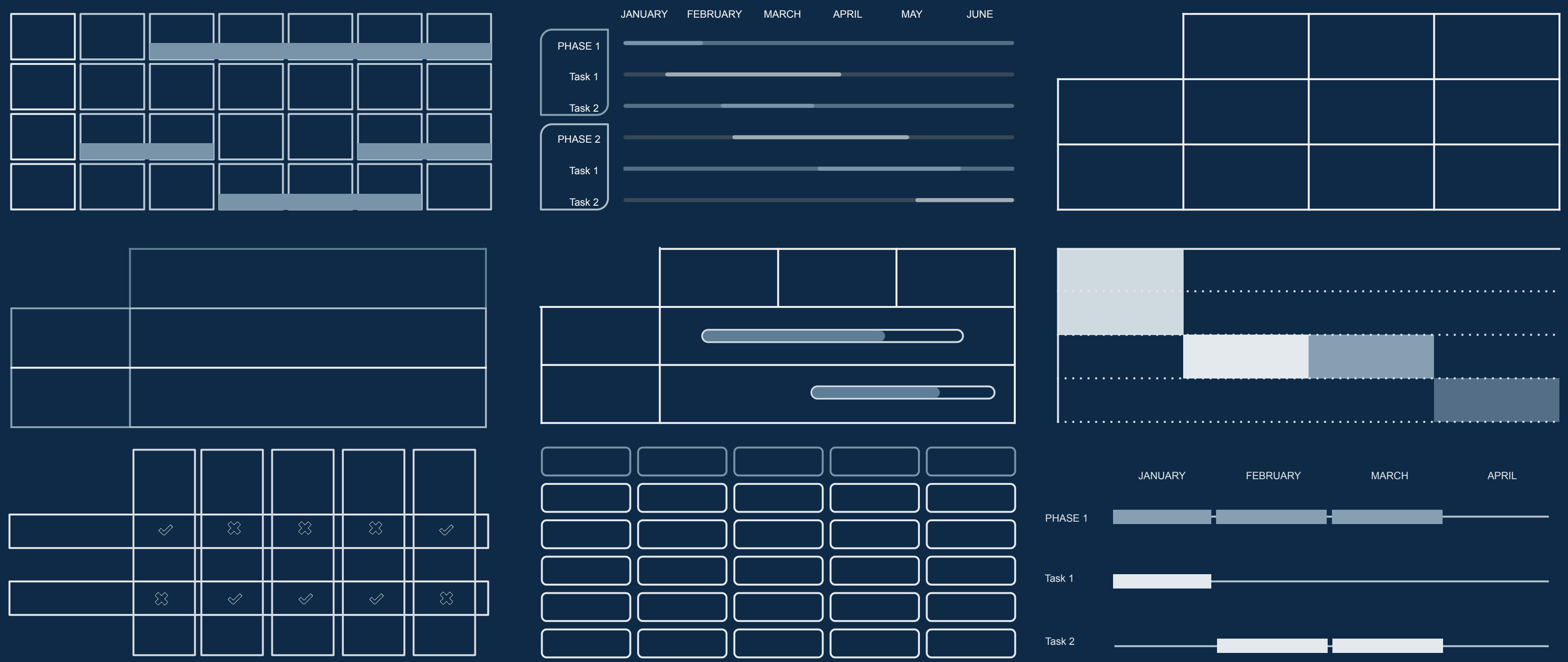
JANUARY
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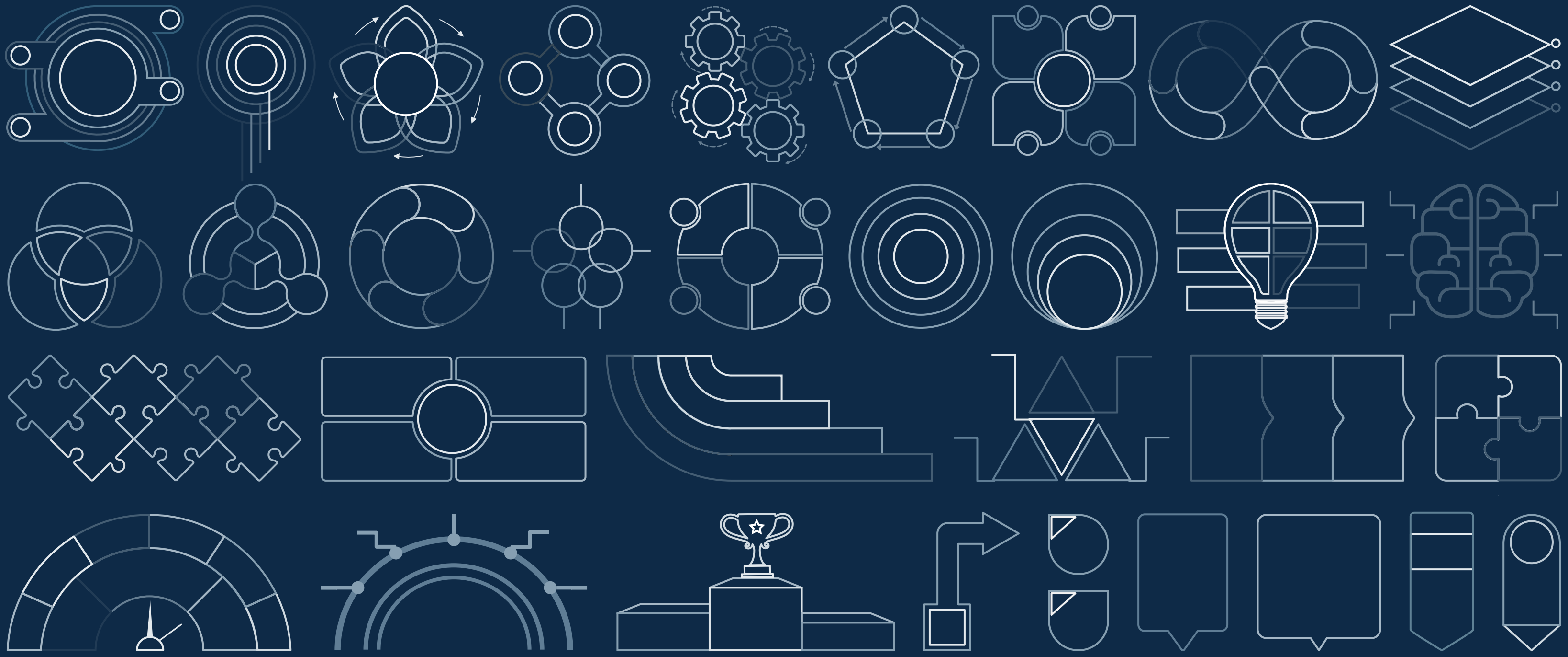

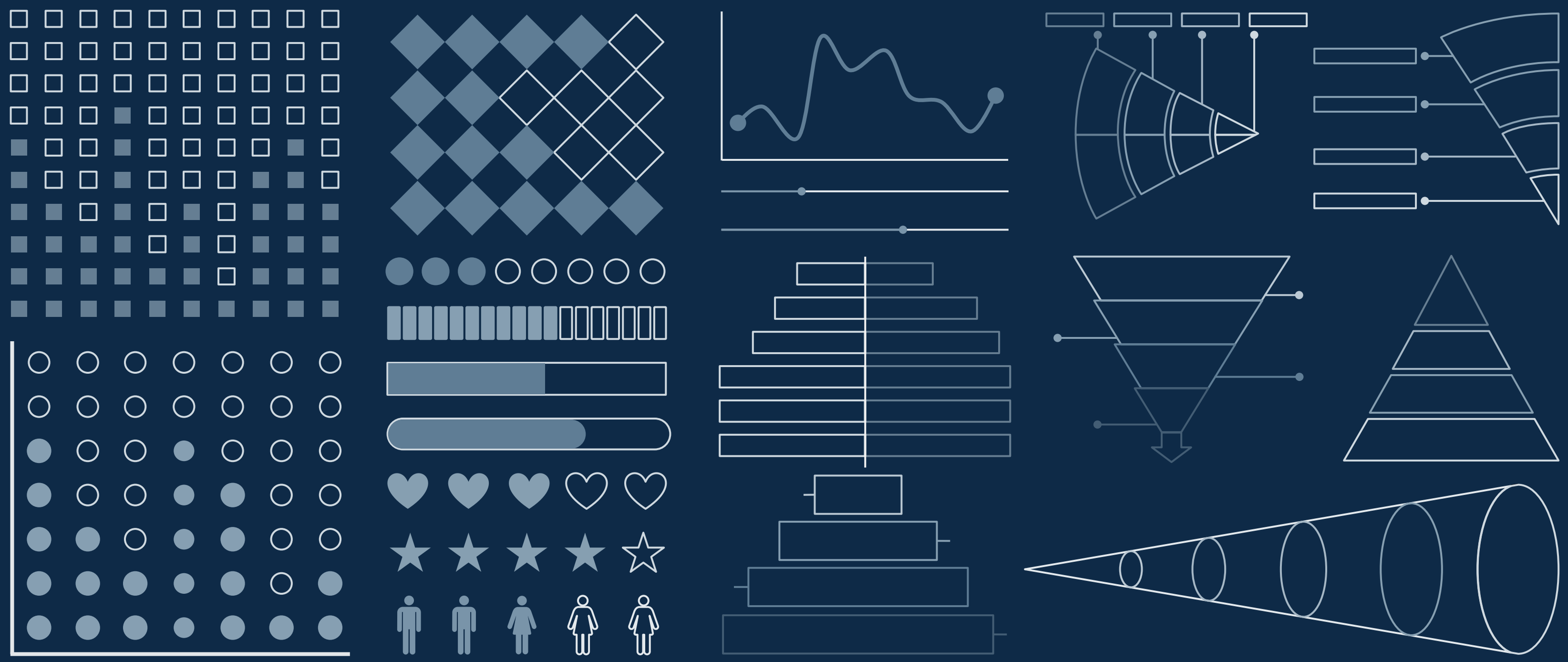

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